Your tumor has a fingerprint no healthy cell has.

Every cancer rewrites its own DNA as it grows.
Those typos exist nowhere else in your body.

That makes them the one thing an immune system could be taught to hunt.

A look at the evidence for one that is actually in trials.

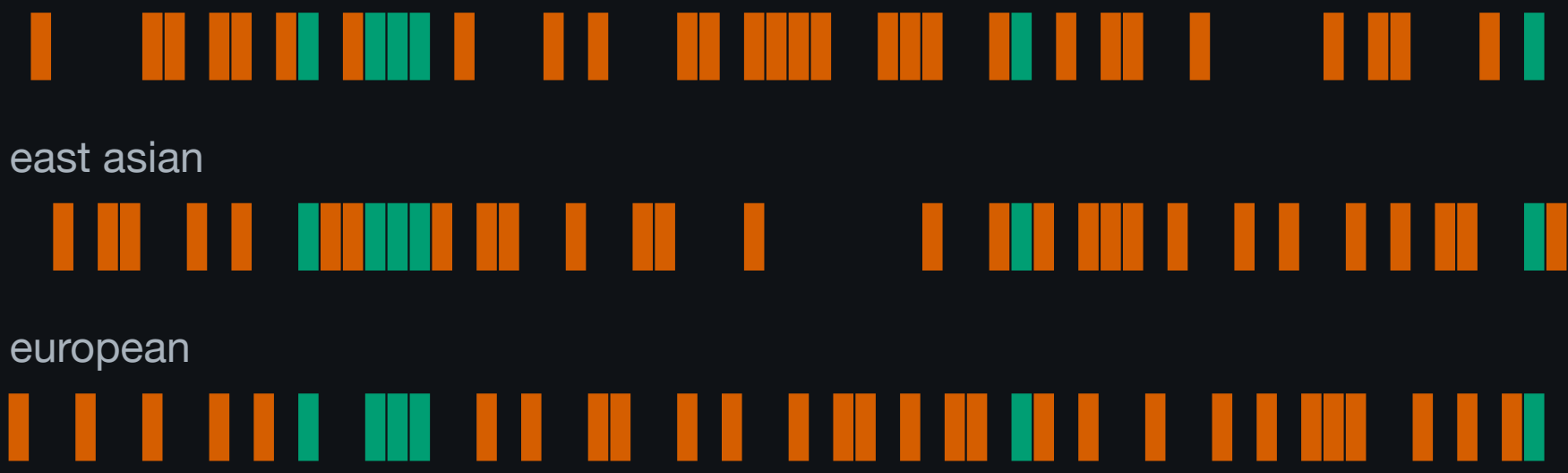
THE IDEA

Turn the typos into wanted posters.

1. Read the tumor's mutations
2. Pick the ones your body can display
3. Write them into one mRNA message
4. Your cells show them to the immune system

A second drug, pembrolizumab, stops the tumor switching those immune cells back off.

Same tumor.
Different person.
Different vaccine.

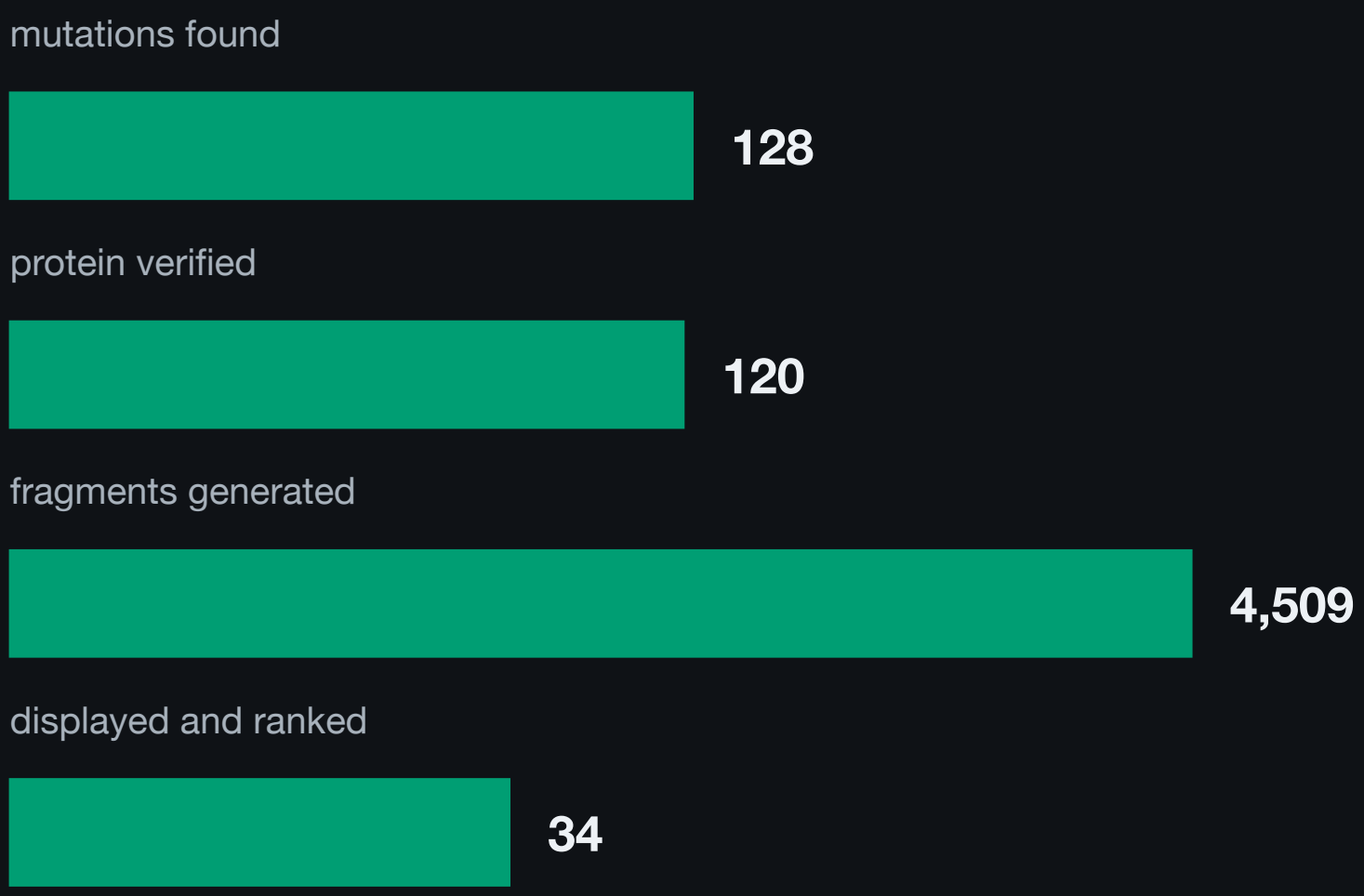


Only 6 of 34 targets are shared.

Which fragments your body can display depends on genes you inherited. Change the person, and about 80% of the vaccine changes with them.

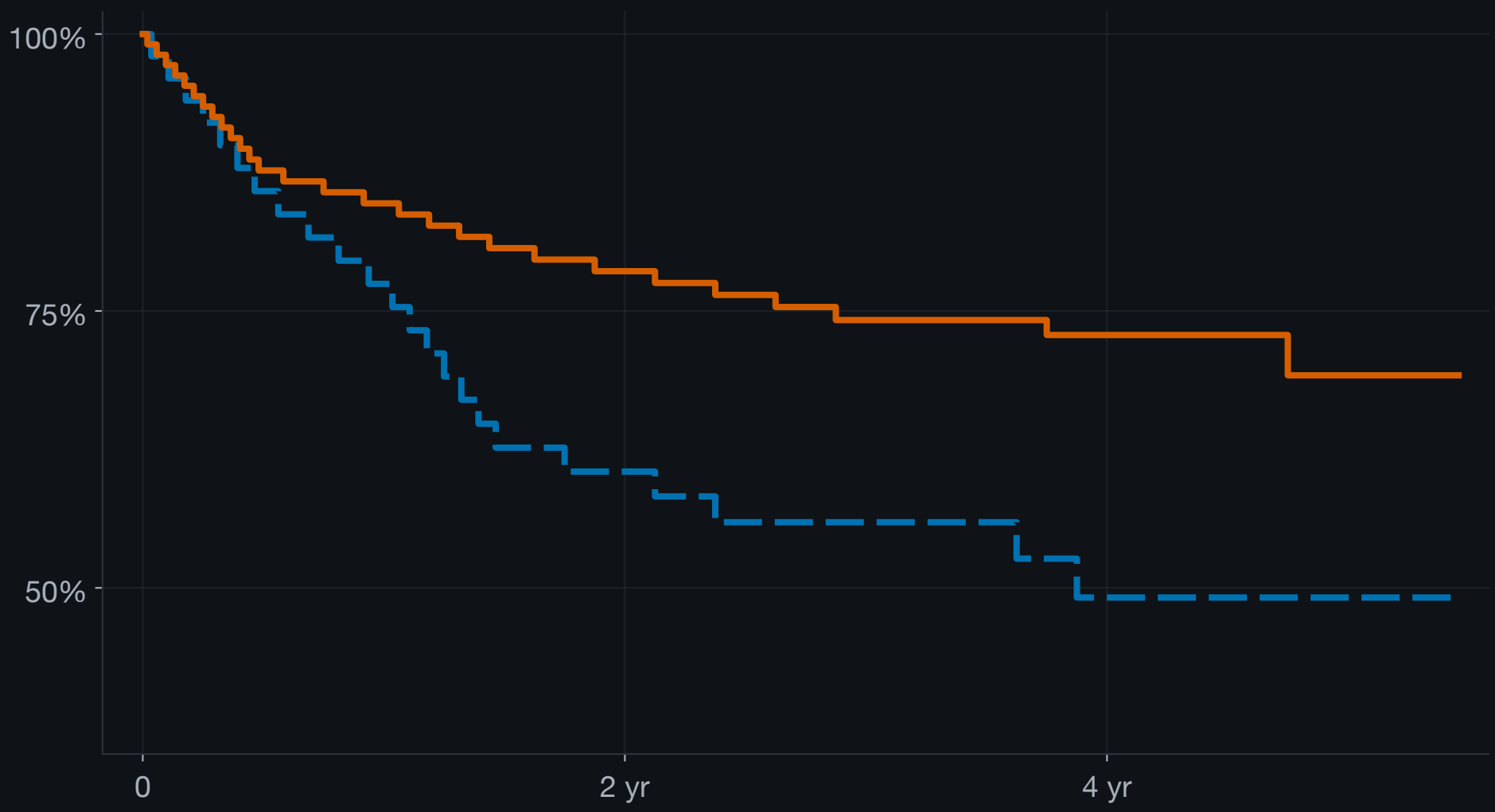
THE NARROWING

From 128 mutations down to 34 targets.



Most candidates are thrown away because they would never be displayed on the cell surface. Every discard is published with its reason.

A 157-person trial, five years of follow-up.



69% vs 49% still cancer-free at 5 years

vaccine + pembrolizumab — pembrolizumab alone

THE SAME RESULT, AS PEOPLE

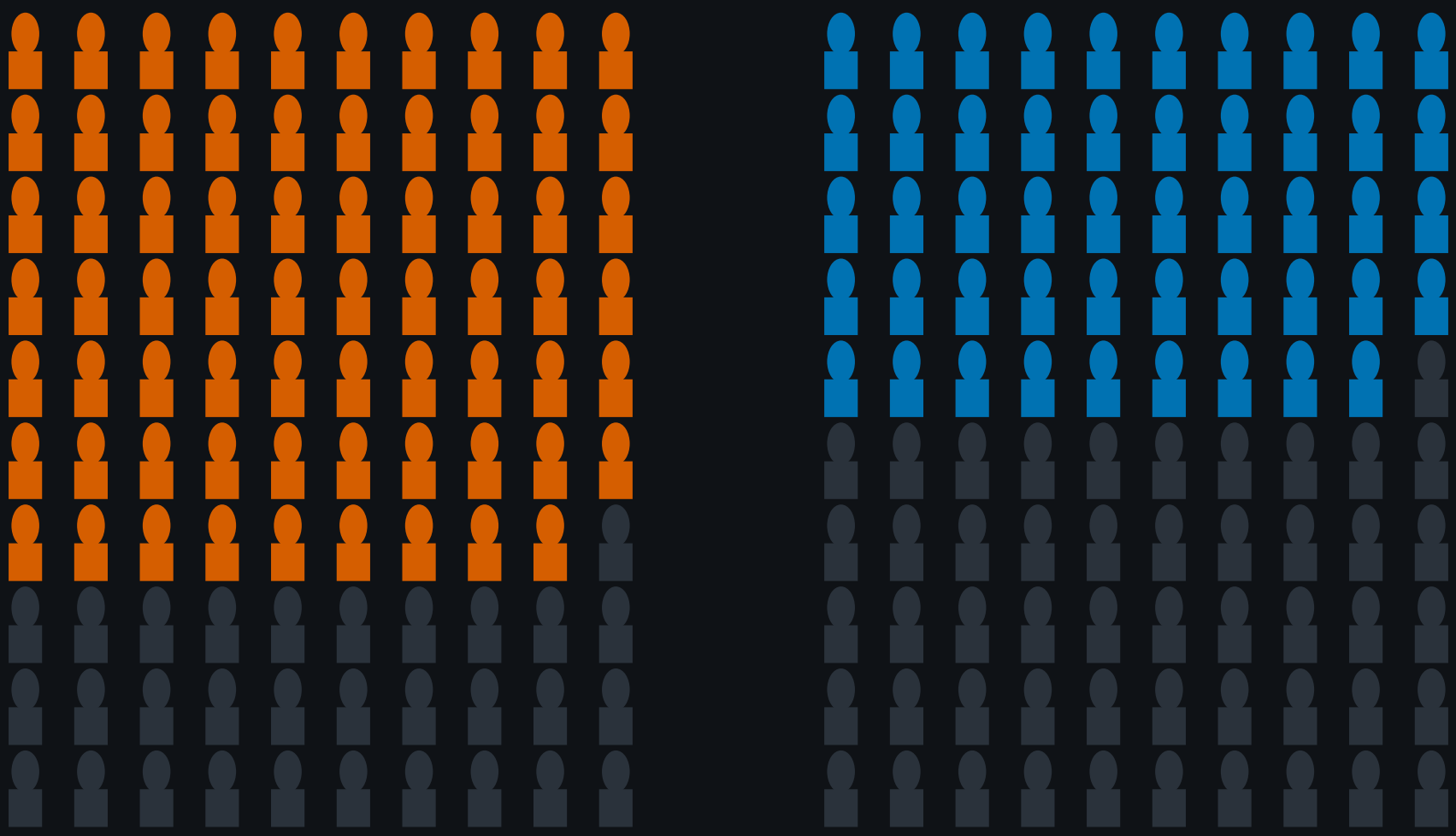
Out of 100 people like this...

69 of 100

vaccine + pembro

49 of 100

pembro alone



20 more people per 100 — but with only 157 patients, the honest range is 1 to 39.

WHAT IT DOES NOT PROVE

Promising.
Not proof.

- 157 patients, 51 recurrences. Small.

- Nobody was blinded — everyone knew who got what.

- The trial's bar was a one-sided p below 0.10. Stated the usual way, it was p = 0.053.

- 14 deaths in total, so it says nothing about living longer.

A much larger blinded trial has reported that it met its goals. No numbers from it are public yet.

HOW TO CHECK THIS YOURSELF

Every number here is reproducible.

The patient-level data was rebuilt from the trial's own published summary table — no curve-tracing, just arithmetic the published table forces.

	reconstructed	published
hazard ratio	**0.506**	0.510
95% interval	**0.291–0.880**	0.294–0.887
recurrences	**28 / 23**	28 / 23

All ten landmark estimates land within half a percentage point of the publication.

OPEN SOURCE

Code, data and sources — all of it.

A reproducible pipeline from a public tumor sample to a conceptual vaccine design, plus an honest re-analysis of the trial evidence.

Every number is sourced with its link, publication date and data cutoff. Anything that could not be verified is flagged and left unused.

github.com/isarinio22/moderna

120 mutations verified · 4,509 fragments scored · 34 selected

Educational project. Not medical advice.
Not affiliated with Moderna or Merck.